

Improving the application of long-term ecology in conservation and land management

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Summary

1. Significant effort is being made to develop more inclusive and systematic decision-making frameworks in ecology, but these have yet to include palaeoecology. Doing so would address critical questions about long-term ecological processes (spanning > 50 years).

2. This paper outlines the main barriers to the integration of long-term ecological data (LTE) into management. Using two UK upland case studies, it uses a choice experiment to assess the value placed on LTE by ecological researchers, policymakers and practitioners. Respondents were able to consider how selecting or excluding different sources of evidence might affect management decisions and their environmental outcomes.

3. The results suggest that LTE has the potential to become a valued part of the evidence base for guiding land-management decisions.

4. *Synthesis and applications.* Placing more emphasis on site-based approaches can help translate this potential into practice by demonstrating the practical benefits of using LTE. By working with managers to address site-based issues, palaeoecology can provide additional insights into ecosystem dynamics and critical thresholds. Using LTE can also improve conservation effectiveness by ensuring that both rapid and lagged responses are anticipated and indicating the range of variability against which management responses can be evaluated.

Key-words: applied palaeoecology, choice experiments, decision-making framework, evidence base, integrative ecology, palaeoecology, stated preferences, upland management

Introduction

The effectiveness of management depends not just on the quality of evidence available to decision-makers, but also on choices made in the decision-making process (Sutherland *et al.* 2004; Mathevet & Mauchamp 2005). Many conservation ecologists recognize the need for decision-making frameworks that accommodate new perspectives and multiple sources of evidence to manage ecosystems for multiple benefits while adapting to uncertainties that lie outside recent experience (Pullin *et al.* 2004; Heller & Zavaleta 2009; Peters 2010). Although progress is being made towards more integrative, multidisciplinary decision-making in some areas (Sutherland *et al.* 2011), this is not the case for all disciplines relevant to conservation management.

Long-term ecological data (LTE; spanning >50 years) has the potential to make significant contributions, but has yet to be routinely recognized in ecological research, policy or practice (Willis & Bhagwat 2010; Rull & Vegas-Vilarrúbia 2011). Davies & Bunting (2010) suggest that answering 54 of the 100 questions of UK conservation importance (Sutherland *et al.* 2006) requires consideration of processes acting over multiple years or of conditions in the past and present. Many palaeoecological papers address themes raised by these 100 questions (Willis *et al.* 2007; Vegas-Vilarrúbia *et al.* 2011), indicating the relevance of long-term data to conservation priorities on an international level. However, only 16% of studies published in the *Journal of Applied Ecology* in 1999 addressed time-scales greater than a decade (Ormerod, Pienkowski & Watkinson 1999), and no key biodiversity assessments published between 1998 and 2005 used records longer than 50 years (Willis *et al.* 2005). The lack of long-term perspectives in ecological policy and research has implications for the effectiveness of management interventions and investment. For example, in marine ecosystems, the

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omission of historical data results in overly optimistic assessments of conservation status, lower recovery targets and higher catch quotas (McClenachan, Ferretti & Baum 2012).

Achieving greater integration of LTE with existing ecosystem management processes is hampered by practical and conceptual barriers and by the shortage of documented examples of practical ways in which LTE could be more effectively integrated into management decision-making. In particular, the lack of opportunities for researchers, policymakers and practitioners to consider how information from different sources might affect management decisions and their environmental outcomes is a significant limitation. Unless the actual and perceived relevance of other sources is assessed, there is a risk that knowledge exchange networks will remain biased towards a subset of established views, rather than accommodating the multiple insights needed to increase resilience (Sutherland *et al.* 2004). This paper uses a choice experiment (CE; Hensher, Rose & Greene 2005) to present multiple dimensions of ecological change and communicate a synthesis of relevant LTE information alongside ecological monitoring and research data. This allows us to assess the willingness of researchers, policymakers and practitioners to incorporate unfamiliar forms of evidence into decision-making, given resource and budget constraints. We also assess the value attributed to stakeholder participation as a means of developing and implementing alternative strategies. We apply the CE to two UK upland ecosystems and use the responses to suggest ways of overcoming barriers to LTE inclusion at the site level, since most palaeoenvironmental literature focuses on potential contributions to ecological theory and implications for management, rather than practical ways of incorporating LTE into management planning and implementation.

Barriers to inclusion

Three interrelated sets of issues limit knowledge exchange between researchers, practitioners and policymakers and restrict integration of ecology and palaeoecology. First, a lack of availability or awareness of relevant long-term and neo-ecological information contributes to differing priorities and a misalignment of interests (Sutherland *et al.* 2009). This is due to insufficient evidence, a shortage of accessible or coordinated data and insufficient communication or evaluation of the effectiveness of existing information (Pullin *et al.* 2004; Willis *et al.* 2007; Newton *et al.* 2009; Davies & Bunting 2010). Second, infrastructural and technical obstacles reduce opportunities for exchange and learning. These include the lack of a support framework and accessible measures for exchange, collation and evaluation of knowledge (Sutherland *et al.* 2004; Newton *et al.* 2009). A lack of time, education and training to provide exposure to relevant ideas from other fields and differences in the ways that various methods record information can further restrict data comparability

(Davies & Bunting 2010). This includes the challenge of translating data into understandable, meaningful formats for other audiences without compromising levels of detail or uncertainty. Finally, attitudes and preconceptions influence the reception of unfamiliar evidence. This includes the perception that longer records are time bound, purely descriptive and of little use in conservation practice (Willis *et al.* 2007), or failure amongst some palaeoecologists to consider conservation ecology as a relevant audience and frame their data accordingly (Birks 2012). It can include uncertainty over data precision or accuracy and reluctance to use data that do not arise from well-controlled experiments (Dietl & Flessa 2011). There may also be cultural resistance to changing established thinking (e.g. Carrion & Fernandez 2009, and responses thereto) or a tendency to focus on the shortcomings of differing sources rather than developing strategies to overcome them (Froyd & Willis 2008).

This paper focuses on attitudes towards LTE and measures for moving towards joint problem-solving. As a starting point for assessing the relative weight that researchers, policymakers and practitioners place on LTE in decision-making, we use a CE as a structured format that allows multiple sources of information to be jointly considered in a manner consistent with decision science (Louviere, Hensher & Swait 2000). CEs have several strengths for assessing preferences towards LTE. First, CEs have been widely used to value ecosystem attributes that are unfamiliar to many stakeholders. This includes assessing how complex concepts like biodiversity are understood and valued by the general public (Christie *et al.* 2006). A variant of the method has been used to assess how unfamiliar evidence, such as the extent of historical woodland change, affects preferences for future changes in tree cover in UK national parks (Hanley *et al.* 2009). Second, CEs take into account the fact that complex decisions are based on multiple decision-relevant criteria by facilitating simultaneous consideration of multiple dimensions of conservation problems, such as changes in raptor populations and local employment under alternative management regimes on sporting estates (N. Hanley, unpublished data). Third, CEs can be used to assess how individuals make trade-offs when multiple, competing benefits and values are involved, as for example in managing ecosystem services (Birol *et al.* 2009). Finally, CEs have advantages over simple attitudinal or ranking surveys since they explicitly reveal the relative value of different sources of information when resources are scarce; that is, they indicate how much people are willing to give up of one type of information to get more of another (given that generating information is costly and that conservation agency budgets are limited), as well as showing which kinds of information conservation professionals view as significant to their choices and management decisions.

Most CEs have been carried out with members of the public to inform policymakers about preferences held by taxpayers, rather than with 'professional' participants

such as ecologists (Burgess, Clark & Harrison 2000) or policymakers (Carlsson, Kataria & Lampi 2011). In contrast, recent efforts to improve policy-making relating to conservation and ecology have involved researchers, practitioners and policymakers in identifying common priorities and emerging issues (Sutherland *et al.* 2011). By involving ecological researchers and practitioners, this paper bridges a gap in CE applications and extends the conservation decision-making literature by assessing the relative merits of longer-term perspectives alongside established sources of evidence and by proposing practical measures for developing more integrative applications of LTE.

Materials and methods

CONSTRUCTION OF A COMPARATIVE EVIDENCE BASE

Choice experiments are a stated preference technique developed in market research, but now used in a range of applications (Bateman *et al.* 2002). Respondents are required to make a series of choices between alternative scenarios to identify their preferences and the trade-offs that they are willing to make between different 'attributes' of a policy option or consumer good. Choices are specified in terms of a number of attributes, each of which is available at different levels. Experimental design consists of selecting attributes and levels and combining them into choice tasks which respondents complete. In this case, participants considered four types of evidence within an upland management context (Table 1):

1. *Ecological monitoring* is used to detect trends and evaluate management effectiveness. Monitoring frequency depends on resources and objectives, including species and ecosystem response rates. Three levels were included in the CE: 3, 6 (the approximate interval in site condition monitoring, the standard approach for monitoring designated sites) and 12 years.

2. *Ecological research* provides the basis for understanding ecosystem behaviour and the underpinning mechanisms, from genome to biosphere scales. Two attribute levels were included in the CE: none (monitoring evidence is sufficient) and diverse (encompassing a broad range of ecological insights, e.g., climate modelling, genetics or carbon chemistry).

3. *Long-term ecological data*: Since many ecosystem processes operate over long periods, baselines may shift between each generation of policymakers, researchers and managers who see only part of the process. This has direct consequences for species and ecosystem management (McClenachan, Ferretti & Baum 2012). 'Long term' here refers to records spanning > 50 years. Three attribute levels were included: none, syntheses (broad-scale) and region- or site-specific data (finer spatial scale).

4. *Stakeholder engagement*: Translating evidence into effective policy and practice requires locally adapted planning and implementation (Heller & Zavaleta 2009). Participation is increasingly advocated as a means of achieving this (Reed 2008) and can improve synergies between research, policy and practice (Sutherland *et al.* 2011). Three attribute levels were included to reflect different levels of participation: none, guidance (stakeholder knowledge or preferences used to implement predetermined strategies) and collaboration (co-generation of research agendas, management or policy).

Choice experiments usually incorporate a price attribute to assess how much participants are willing to pay to maintain particular landscape characteristics or to support a change in management, for example. Time is included as a fifth attribute to represent the cost of changing or broadening the evidence base used to support management decisions, since time is a scarce resource for conservation managers. These costs are prospectively incurred in acquiring new information, through the time taken to gain a basic understanding of additional sources, keep abreast of developments in a wider range of fields, or take part in meetings or projects in order to obtain additional types of evidence. Three time costs are included in the CE: 1 day quarter⁻¹, 1 day month⁻¹ and 1 day week⁻¹. This relates to how much working time would be allocated within a participant's organization, rather than at the level of the individual respondent.

To provide a real-world context, information on each attribute relating to two practical UK conservation issues was presented to participants before they were asked to complete the choice cards: the management of upland peatlands and upland woodlands, in each case with the aim of maintaining ecosystem viability. These contexts were selected because their management incorporates a range of biotic and abiotic interactions. They also include values arising from a complex palimpsest of environmental changes and cultural activities on recent to millennial time-scales (Tallis 1998), with scope for broad disciplinary and knowledge input. They are sensitive to climatic and management change and provide many ecosystem services, with beneficiaries well beyond the geographical limits of the uplands (Holden *et al.* 2007; Reed *et al.* 2009). Reliance on partial information for making management decisions has potentially significant and widespread implications for the long-term supply of these ecosystem services.

In applying a CE to evaluate how information provision influences participant preferences, the clarity, relevance and acceptability of that evidence is paramount. Therefore, information on each attribute was drawn from peer-reviewed literature and best practice guidance (the information presented to respondents is available on request from the corresponding author). The information aimed to summarize the current state of knowledge, with a focus on ecosystem process and function, in the context of key management issues (Holden *et al.* 2007; Hopkins & Kirby 2007; Sutherland *et al.* 2010).

Table 1. A sample 'choice card' showing how different levels of attributes were combined to provide three hypothetical alternatives from which participants were asked to select their most preferred (or least disliked) option

Source	Alternative 1	Alternative 2	Alternative 3
Ecological monitoring	Every 12 years	Every 6 years	Every 3 years
Ecological research	None	Diverse	Core
Long-term research	None	General	Specific
Stakeholder preferences	None	Bottom-up	Top-down
Time commitment	1 day 4 months ⁻¹	1 day month ⁻¹	1 day week ⁻¹
Choice			

EXPERIMENTAL DESIGN AND IMPLEMENTATION

The CE was designed and implemented in two stages. A pilot survey was conducted with a small number of participants to improve the statistical efficiency and ease of use of the final design. This included questions on the amount, relevance and clarity of information presented. A D-optimal design (Rose & Bliemer 2009) was used to combine the attributes offered to respondents. This requires explicit incorporation of prior information about respondents' preferences, which was obtained from the mean and standard deviations of the estimated attribute coefficients from the pilot survey, on the assumption that respondent preferences for the full survey sample will lie within this range. Coefficients that were not significant in the pilot survey were assigned a fixed zero value. Participants were asked to select the context (woodland or peatland) with which they were most familiar before completing the CE. Each participant completed 18 choice cards.

The survey was completed by professionals with experience of the habitats and issues described, including representatives from government agencies, non-governmental organizations, researchers and practitioners. Participants were recruited via personal contacts, email invitations to members of UK upland policy and research networks, and additional participants suggested by these respondents, so no response rate is available. The survey was conducted online. All responses were treated anonymously.

Results

Sixteen completed responses were received, including one NGO ecologist, one policymaker, three practitioners, five researchers and six agency ecologists, drawn from England, Wales, Scotland and the Irish Republic.

A random parameter logit model using normally distributed preferences provided the best fit for both CEs. In the peatland CE (Table 2), for the non-random parameters in each respondents' preference, ecological research is significantly valued relative to no such input to decision-making, as is LTE at both 'synthesis' and 'specific' levels, with a slight preference for the former. Time commitments to information processing are not significant determinants of choice. Preferences for ecological monitoring at 3-year intervals do not differ from those at 6-year intervals, but participants respond negatively to a change to 12-year intervals. Both stakeholder guidance and collaboration are valued relative to no such involvement, with a higher value placed on collaboration than guidance. Preferences vary significantly across respondents for ecological monitoring and stakeholder inputs, as standard deviation estimates are strongly significant.

The model fit for the woodland case study (Table 3) is not as strong as that for the peatland CE [pseudo R^2 of 0.38 relative to 0.52: values of 0.3–0.4 are equivalent to R^2 of 0.6–0.8 in standard linear models (Hensher, Rose & Greene 2005)]. In contrast with the peatland CE, there is no significant preference for ecological monitoring intervals differing from a 6-year frequency, since parameter estimates for 3- and 12-year intervals are insignificant. On average, respondents consider ecological research to

Table 2. A model for peatland choice experiment responses showing coefficient estimates for each attribute and their associated standard errors (SE), with standard deviation estimates (SD) for attributes modelled as randomly distributed across respondents

Attribute	Coefficient	SE
Random parameters in utility functions		
Ecological monitoring 3 years (relative to 6 years)	0.668	0.553
Ecological monitoring 12 years (relative to 6 years)	−4.439***	1.117
Stakeholder guidance (relative to none)	2.719***	0.747
Stakeholder collaboration (relative to none)	4.639***	1.310
Non-random parameters in utility functions		
Ecological research (relative to none)	4.132***	1.093
LT research synthesis (relative to none)	4.087***	1.098
LT research specific (relative to none)	3.829***	1.097
Time 1 day month ^{−1} (relative to 1 day quarter ^{−1})	−0.443	0.463
Time 1 day week ^{−1} (relative to 1 day quarter ^{−1})	0.287	0.378
Attribute	Coefficient	SD
Distributions of random parameters (standard deviation estimates)		
Ecological monitoring 3 years	4.054***	1.139
Ecological monitoring 12 years	4.832***	1.407
Stakeholder preference guidance	2.163***	0.675
Stakeholder preference collaboration	3.523***	1.098
Log likelihood at convergence	−84.91	
Pseudo R^2	0.52 (52%)	

*** Denote significance at 1% level.

be valuable. Both synthesis-level and more specific LTE data are preferred to none. The larger coefficient for specific LTE data (1.59) reveals that it is slightly preferred over synthesis data (1.36). Stakeholder involvement is preferred relative to none, with collaboration preferred (0.86) over guidance (0.69). The lowest time demand (once per 4 months) is preferred over monthly or weekly time inputs, and unlike the peatland CE, both these measures of time demand are of significance to respondents, as shown by statistically significant negative parameter estimates for 1 day month^{−1} and 1 day week^{−1}. There is statistically significant heterogeneity in values attached to the frequency of ecological monitoring and to ecological research as inputs to management decision-making.

Discussion

This study assesses the value placed on LTE by ecological researchers, policymakers and practitioners, using a CE as a vehicle for presenting these unfamiliar data alongside

Table 3. A model for woodland choice experiment responses, showing coefficient estimates for each attribute and their associated standard errors (SE), with standard deviation estimates (SD) for attributes modelled as randomly distributed across respondents

Attribute	Coefficient	SE
Random parameters in utility functions		
Ecological monitoring 3 years (relative to 6 years)	-0.066	0.550
Ecological monitoring 12 years (relative to 6 years)	-0.582	0.512
Ecological research (relative to none)	2.356***	0.655
Non-random parameters in utility functions		
LT research synthesis (relative to none)	1.370***	0.421
LT research specific (relative to none)	1.592***	0.503
Stakeholder guidance (relative to none)	0.692***	0.339
Stakeholder collaboration (relative to none)	0.868***	0.342
Time 1 day month ⁻¹ (relative to 1 day quarter ⁻¹)	-0.534*	0.308
Time 1 day week ⁻¹ (relative to 1 day quarter ⁻¹)	-1.254***	0.329
Attribute	Coefficient	SD
Distributions of random parameters (standard deviation estimates)		
Ecological monitoring 3 years	1.257***	0.487
Ecological monitoring 12 years	0.798**	0.417
Ecological research	1.177**	0.616
Log likelihood at convergence	-98.79	
Pseudo R ²	0.38 (38%)	

***, **, * Denote significance at 1%, 5% and 10% levels, respectively.

established approaches for making management decisions. The CE results are discussed before suggesting ways of improving the practical applicability of LTE for managers working at the site level.

PREFERENCES TOWARDS LONG-TERM ECOLOGY IN THE UPLAND EVIDENCE BASE

The CE results provide the first quantitative indication that ecologists see a potential value in palaeoenvironmental data as an additional source of information for making management decisions. A preference for detailed, rather than broad-scale, long-term woodland data may reflect the supporting evidence presented to participants: regional variations in the timing and extent of range shifts are more pronounced for UK woodlands (Tipping 1994) than peatlands (Tallis 1998). This preference may also reflect current management concerns, such as the continuing contraction of old-growth woodland (Hopkins & Kirby 2007) compared with the relative stability of moorland extent since the 1990s (Carey *et al.* 2008). Similarly, more rapid response rates in peatland species, compared

with slower growth rates and generation times for trees, may make 12-year ecological monitoring intervals less desirable than 3- or 6-year intervals on peatlands, compared to no statistically significant preferences for a change from current 6-year intervals in woodlands.

In terms of costs, for peatlands, time costs within a participant's organization were not viewed as a significant factor. This suggests that changes are not seen as significantly different from current requirements or that it is difficult to estimate on an organizational rather than individual level, leading participants to overlook this attribute when making their choices. For woodlands, the lowest time demand (1 day quarter⁻¹) was preferred to higher levels, indicating that participants clearly viewed an increase to 1 day week⁻¹ as a significant and undesirable burden. Any changes to existing working practice to incorporate LTE must acknowledge these constraints.

While the small sample size and lack of opportunities for participants to query the data set or discuss reasons for their choices make it difficult to draw wider inferences from the data, the consistency of the responses across a range of participant positions (government agency and policy, academics and practitioners) suggests that these results could have broader relevance and applicability, at least within UK upland management.

BUILDING AN INTEGRATIVE APPROACH

It may be intuitively unsurprising that participants express a preference for using a range of evidence to inform management decisions, but the CE provided a rare opportunity for them to consider the implications of selecting or omitting different sources. Although the CE participants considered LTE useful for decision-making, palaeoecology remains absent from applied management and priority-setting exercises (Sutherland *et al.* 2009, 2010). This suggests that current strategies for increasing awareness and application of palaeoecology in management are not widely effective. Using the CE responses, practical applications are outlined to help bring about closer integration.

Collaboration between ecologists and managers can increase the effectiveness of management on the ground, in contrast with academic papers that often have limited impact on daily management decisions (Caudron, Vigier & Champigneulle 2012). This is reflected in the preference for collaborative management expressed in the CE. Using palaeo-methods that can readily be combined with modern monitoring measures provides a starting point for collaboration between palaeoecologists, ecologists and managers. For example, measured and diatom-inferred pH can be combined with palaeo-diatom records to monitor water quality and track recovery from historic acidification for implementing the Water Framework Directive (Battarbee *et al.* 2012). Similarly, testate amoebae and organic matter in surface and old peat could be used to monitor water-tables and peat accumulation during

peatland restoration, thus linking restoration progress to longer-term ecosystem processes (Davis & Wilkinson 2004; Laggoun-Défarge *et al.* 2008). Relatively familiar methods can therefore provide a bridge between contemporary and LTE.

Direct legislative impetus for interaction between managers and palaeoecologists, such as the Water Framework Directive, is rare, so establishing clear mutual interests may help motivate and catalyse collaboration (cf. Caudron, Vigier & Champigneulle 2012; Sergeant, Moynahan & Johnson 2012). Using participatory methods to identify shared priorities (Sutherland *et al.* 2011) can help clarify issues, identify gaps in knowledge and improve the speed of implementation (Davies & Bunting 2010; Rudd 2011). This approach has recently been applied in palaeoecology to identify the 50 most pressing questions relevant to key ecological issues. Similar efforts amongst palaeoecologists, ecologists and managers would help palaeoecologists to present long-term messages in ways that are more relevant to real-world, practical management and policy issues, and encourage mutual learning and adaptation (Dietl & Flessa 2011; Birks 2012). Collaboration between palaeoecologists and organizations that own or manage multiple sites could help identify common LTE and management interests beyond a single-site level; this could also focus attention on how common priorities relate to real-world problems (cf. Sayer *et al.* 2012). Based on the CE results, shared priorities might include thematic topics such as biodiversity (Willis *et al.* 2007) as well as habitat-specific issues (Chambers *et al.* 2007).

The strongest contributions from LTE to conservation management include understanding ecological processes, establishing management baselines and identifying critical ecosystem thresholds (Willis *et al.* 2005). Conservation planning would therefore benefit if long-term perspectives were considered early in the decision-making process. Working with ecologists and managers to develop applied research programmes that incorporate LTE into existing ecological monitoring and survey networks (e.g. designated sites, UK Environmental Change Network) would allow LTE to be considered in early-stage decisions. Using this approach, site-specific management baselines could be developed and assessed from multiple perspectives. For instance, by designing LTE to address site-specific lake management questions set by managers, palaeolimnologists were able to evaluate proposed management activities and make practical recommendations to improve their conservation effectiveness (Sayer *et al.* 2012). This can result in multi-use solutions, such as identifying species conservation strategies that also maintain recreational fishing values (Sayer *et al.* 2012) or allowing managers to maintain conservation values and public access, rather than considering costly mimicry of historic management systems (Waller 2010).

Using LTE within existing ecological frameworks would increase impact beyond single sites and incur a lower time demand than setting up new networks. In addition,

linking LTE with the multidecadal trends that are emerging from established monitoring networks could help identify delayed responses that can prevent timely management adaptation (Monteith & Evans 2005; Morecroft *et al.* 2009; Youngblood & Palik 2011; Hughes *et al.* 2012). This would also provide opportunities to develop new methods of data comparison and integration (Froyd & Willis 2008; Hanley *et al.* 2008; Sayer *et al.* 2010; Davies 2011; Birks 2012), particularly where data bases are available to help test new cross-scale methods more widely (Polly *et al.* 2011). Undoubtedly this will require compromise as no sites or data sets will be perfect where multiple interests are involved (Birks 2012).

CONCLUSIONS

Examples where palaeoecology is used to help make practical management decisions are rare, and it is unclear how these long-term messages are received by ecologists, managers or policymakers. The results of the CE presented in this paper provide the first quantitative evidence that participating researchers, practitioners and policy advisers see a potential value in using long-term ecology as part of the evidence base for UK upland management. We argue that placing more emphasis on site-based approaches can help translate this potential into practice by demonstrating the practical benefit of using LTE to managers. This includes introducing LTE as an additional monitoring tool that links short-term responses to longer-term ecosystem processes and using LTE to help improve the conservation effectiveness of proposed or existing management strategies. Articles in journals and practitioners' magazines remain essential for raising awareness of the links between LTE and management, but literature documenting efforts at integration is currently rare (Sayer *et al.* 2012); by helping to communicate lessons learnt by palaeoecologists, ecologists and managers, well-documented examples of collaboration could make future efforts more effective.

The mounting pressures on ecosystems and time delays between information gathering, policy development and implementation mean that continuing to overlook information on longer-term ecosystem function and process represents a growing risk to the future effectiveness of management decisions. Working to incorporate LTE alongside established ecological and stakeholder evidence can allow us to better anticipate the emergence of individualistic or novel responses and allow the timely identification of critical thresholds to help manage and maintain ecosystem function and services over the longer term.

Acknowledgements

This research was funded by the Economic and Social Research Council through the Rural Economy and Land-Use Programme (RES-229-27-0003). Thanks to choice experiment participants for their assistance and to Des Thompson and the referees for providing useful improvements to the paper.

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Received 13 March 2013; accepted 2 August 2013

Handling Editor: Des Thompson